

Name: _____



Penrith High School

2013

Trial Higher School
Certificate Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the end of this paper.
- A multiple choice answer sheet is provided at the end of this paper.
- Show all necessary working for questions 11 – 16 in booklets.
- Work on this question paper will not be marked

Total marks – 100

Section 1 – 10 marks

- Attempt questions 1 – 10
- Allow about 15 minutes for this section.

Section 2 – 90 marks

- Attempt questions 11 – 16
- Allow about 2 hours and 45 minutes for this section.

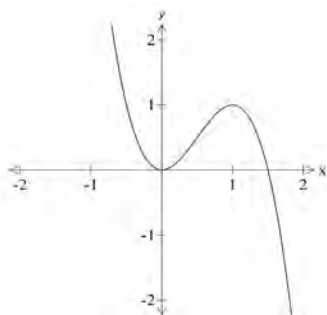
Questions		
1, 2, 3, 11 α		/12
4, 11 β		/7
5, 6		/2
7, 8, 15 α		/7
9, 10, 13		/17
12		/15
14		/15
15 β		/10
16 α		/4
16 β		/11
Total		/100

Section 1

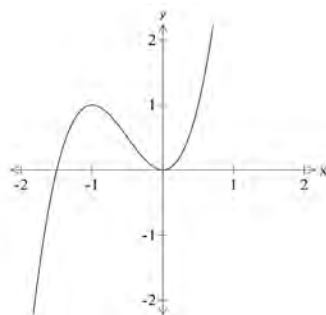
1. Factorise $3x^2 + 9x$
(A) $3x(x+3)$ (B) $3x(x+9)$ (C) $3x(x+12)$ (D) $3x(3x+3)$
2. $\frac{1}{4} + \frac{x}{8} = ?$
(A) $\frac{1+x}{8}$ (B) $\frac{2+x}{8}$ (C) $\frac{1+x}{12}$ (D) $\frac{8+x}{32}$
3. What is the simultaneous solution to the equations $x+y=1$ and $3x-y=7$?
(A) $x=2$ and $y=-1$ (B) $x=3$ and $y=-2$
(C) $x=2$ and $y=3$ (D) $x=3$ and $y=4$
4. What is the centre and radius of the circle with the equation $x^2 + y^2 + 6x - 8y - 11 = 0$?
(A) Centre $(-3, -4)$ and radius 36 (B) Centre $(-3, 4)$ and radius 36
(C) Centre $(-3, -4)$ and radius 6 (D) Centre $(-3, 4)$ and radius 6
5. The chance of a fisherman catching a legal length fish is 4 in 5. If three fish are caught at random, what is the probability that exactly one is of legal length?
(A) $\frac{4}{125}$ (B) $\frac{12}{125}$ (C) $\frac{16}{125}$ (D) $\frac{48}{125}$
6. Sixty tickets are sold in a raffle. There are two prizes. Lincoln buys 5 tickets. Which expression gives the probability that Lincoln wins both prizes?
(A) $\frac{5}{60} + \frac{4}{59}$ (B) $\frac{5}{60} + \frac{4}{60}$ (C) $\frac{5}{60} \times \frac{4}{59}$ (D) $\frac{5}{60} \times \frac{4}{60}$
7. What are the x-coordinates of the two stationary points to the curve $y = 5 + 3x^3 - 2x^4$?
(A) $x=0, x=\frac{2}{3}$ (B) $x=0, x=\frac{3}{2}$ (C) $x=0, x=\frac{8}{9}$ (D) $x=0, x=\frac{9}{8}$

8. Which of the following is the graph of $f(x) = 2x^3 - 3x^2$?

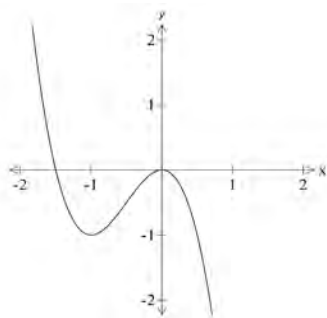
(A)



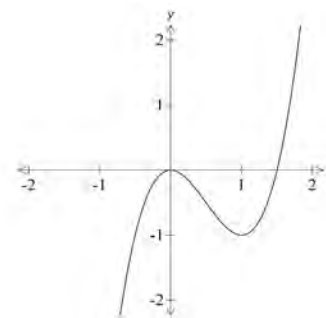
(B)



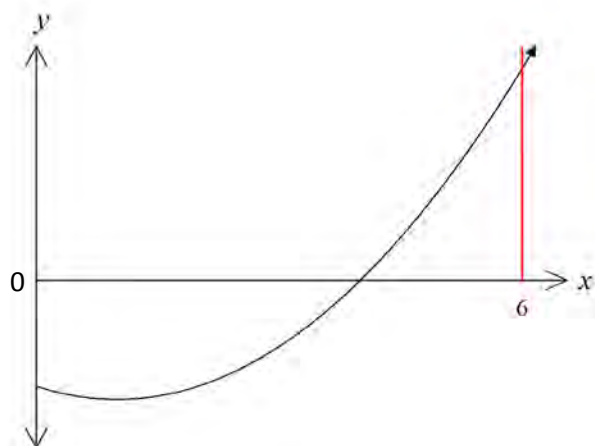
(C)



(D)



9. The diagram below shows the graph of $y = x^2 - 2x - 8$.



What is the correct expression for the area bounded by the x -axis and the curve $y = x^2 - 2x - 8$ between $0 \leq x \leq 6$?

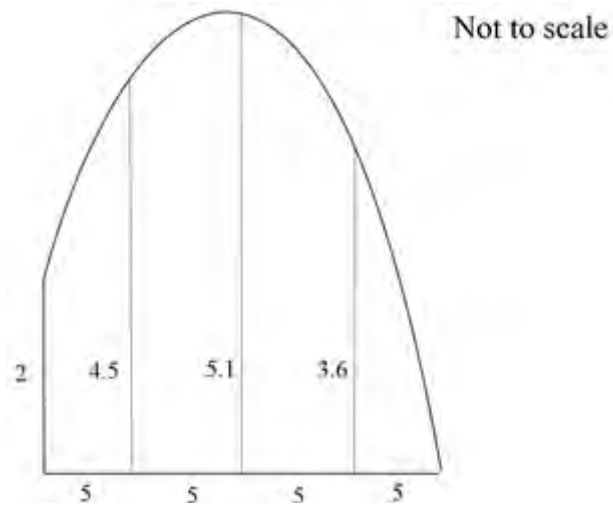
(A) $A = \int_0^5 x^2 - 2x - 8 dx + \left| \int_5^6 x^2 - 2x - 8 dx \right|$

(B) $A = \int_0^4 x^2 - 2x - 8 dx + \left| \int_4^6 x^2 - 2x - 8 dx \right|$

(C) $A = \left| \int_0^5 x^2 - 2x - 8 dx \right| + \int_5^6 x^2 - 2x - 8 dx$

(D) $A = \left| \int_0^4 x^2 - 2x - 8 dx \right| + \int_4^6 x^2 - 2x - 8 dx$

10. The diagram below shows a native garden. All measurements are in metres.



What is an approximate value for the area of the native garden using the trapezoidal rule with 4 intervals?

- (A) 31 m^2 (B) 62 m^2 (C) 71 m^2 (D) 74 m^2

Section 2: Free response

Start each question in a new booklet.

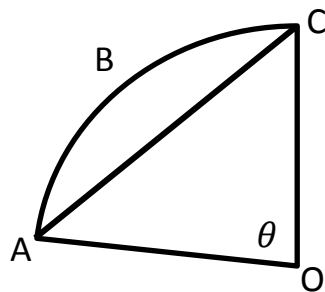
Question 11 Start this question in a new booklet.

Marks

- | | | | |
|------------|------|--|---|
| α) | i. | Evaluate to 1 decimal place: $(1 \cdot 2)^2 + (2 \cdot 1)^2$ | 1 |
| | ii. | Simplify $5\sqrt{3} + \sqrt{20} - 2\sqrt{12}$ | 2 |
| | iii. | Factorise $3x^2 + 8x - 16$ | 2 |
| | iv. | Solve for n : $ 3n - 5 = 4n - 7$ | 2 |
| | v. | Express as a single fraction in simplest form: $\frac{(3x-4)}{2} - \frac{(x+1)}{5}$ | 2 |
| β) | i. | If α and β are the roots of $3x^2 - 6x + 12 = 0$, find the value of: | |
| | a. | $\alpha + \beta$ | 1 |
| | b. | $\alpha^2 + \beta^2$ | 2 |
| | ii. | For the equation $x^2 + (m - 3)x + m = 0$ | |
| | a. | Find an expression (in simplest form) for the discriminant $b^2 - 4ac$, in terms of m . | 1 |
| | b. | Hence find the values of m for which the equation has two roots. | 2 |

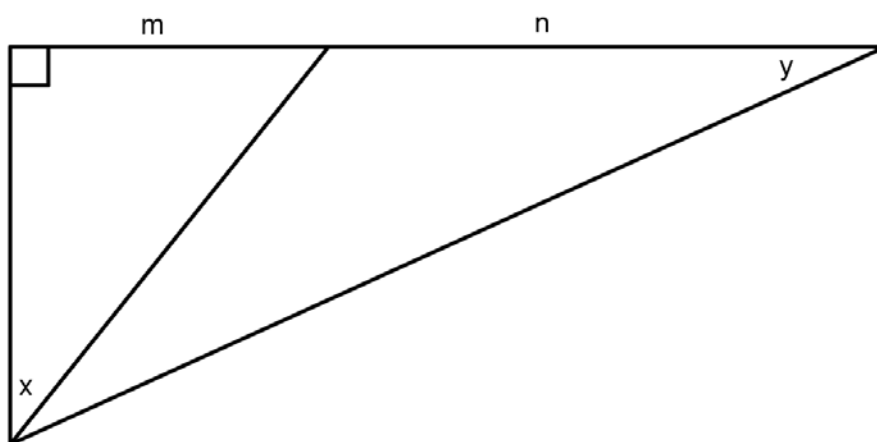
Question 12 Start this question in a new booklet

- i. Find the exact value of $\cos \frac{\pi}{6}$ 1
- ii. Solve $5\cos^2\theta + 2\sin\theta - 2 = 0$ for $0 \leq \theta \leq 2\pi$ 2
- iii. Sketch $y = 2\cos 3x$ for $0 \leq x \leq 2\pi$. 2
- iv. Differentiate $\sin 2x$ 1
- v. Prove that $\frac{\sin\theta\cos^2\theta - \sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$ 2
- vi. The diagram shows a sector OAC with area $90\pi\text{cm}^2$ and $OA = 15\text{ cm}$.



Not to scale

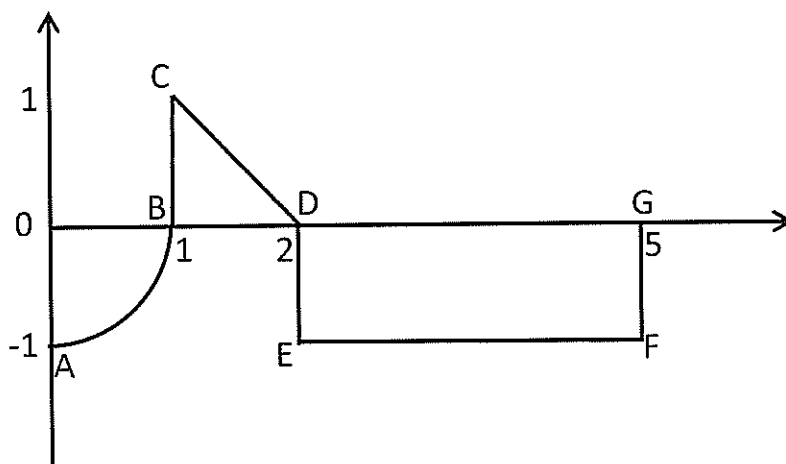
- a. Find the size of θ in radians. 2
- b. Find the perimeter of the segment ABC. Give your answer correct to the nearest cm . 3
- vii. In the diagram, prove that $\tan x \tan y = \frac{m}{m+n}$ 2



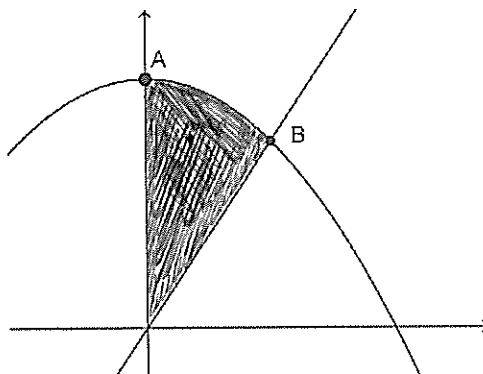
Question 13 Start this question in a new booklet

- i. The gradient of a curve is given by $\frac{dy}{dx} = 6x^2 - 6$. The curve passes through the point (2, -1). Find the equation of the curve. 2
- ii. By first constructing a table of values, use Simpson's rule with 5 function values to approximate the area enclosed between the curve $y = \frac{1}{5-x}$, the x -axis and the lines $x = 0$ and $x = 4$. Answer correct to 2 significant figures. 3

- iii. A function, $y = f(x)$, is drawn below and represented by the curve ABCDEFG. Note: AB is an arc of a circle, centre (0, 0) and radius 1 unit.



- a. Show that the area enclosed between the curve $y = f(x)$, the x axis and the ordinates $x = 0$ and $x = 5$ is $\frac{\pi+14}{4}$. 2
- b. For the above information, find the value of $\int_0^5 f(x)dx$.
Leave your answer in terms of π . 2
- iv. In the diagram, the shaded region is bounded by the parabola $y = -x^2 + 1$, the y -axis and the line $y = \frac{3x}{2}$. 1
- a. Find A, the y -intercept of the parabola $y = -x^2 + 1$. 2
- b. Find B, the point of intersection of the parabola and the line. 3
- c. Find the volume of the solid formed when the shaded region is rotated about the y -axis.



Question 14 Start this question in a new booklet.

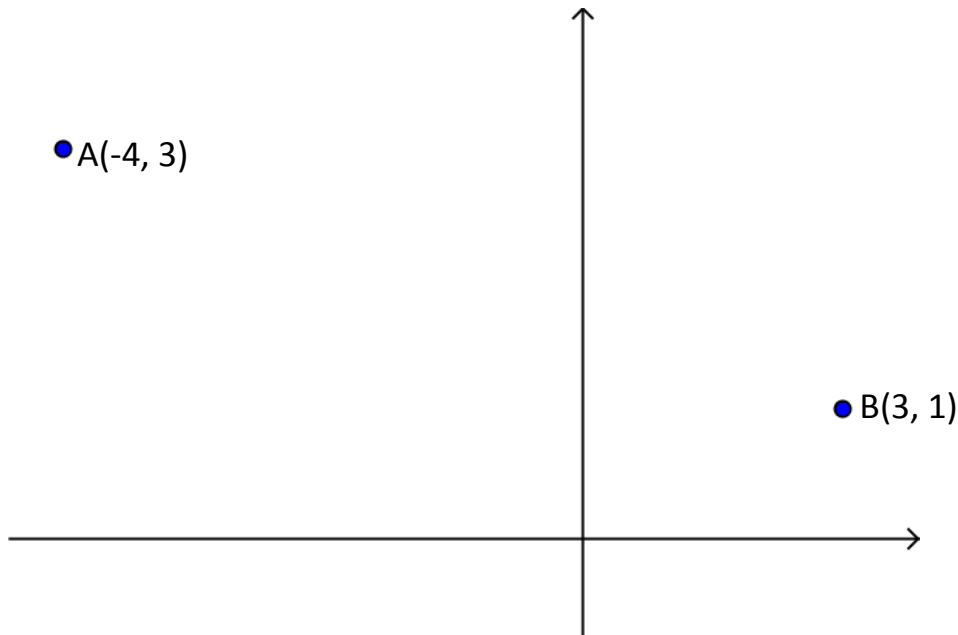
- i. Solve $3\log_7 2 = \log_7 x - \log_7 4$ 2
- ii. Differentiate the following with respect to x :
a. $x^2 \log_e x$ 2
b. $\frac{e^x}{3x-2}$ 2
- iii. Find $\int 5e^{2x} dx$ 1
- iv. Find $\int \frac{4}{2x-7} dx$ 2
- v. Given that $\frac{d}{dx}(e^{x^2}) = 2xe^{x^2}$, evaluate $\int_0^1 xe^{x^2} dx$. 2
- vi. Find the equation of the tangent to the curve $y = 2xe^x$ at the point $(1, e)$. 2
- vii. Find the area enclosed by the curve $y = \log_e x$, the x axis and the line $x = 2$. 2

Question 15 Start this question in a new booklet.

- α) Consider the curve given by $y = x^3 - 3x^2 - 9x + 1$.
- i. Find $\frac{dy}{dx}$ 1
 - ii. Find the coordinates of the two stationary points. 2
 - iii. Determine the nature of the stationary points. 1
 - iv. Sketch the curve clearly showing stationary points and y intercept. 1
- β)
- i. A particle moves in a straight line so that its displacement, x metres from a fixed point on a line, is given by $x = t + \frac{25}{t+2}$ where t is measured in seconds.
 - a. Find the particle's initial position. 1
 - b. Find expressions for the velocity and acceleration in terms of t . 2
 - c. Find when and where the particle is at rest. 2
 - d. Find the limiting velocity of the particle. 1
 - ii. The number N of bacteria in a culture at a time t seconds is given by the equation $N = 25\,000e^{0.007t}$
 - a. What is the number of bacteria initially? 1
 - b. Determine the number of bacteria after 30 seconds (to the nearest whole number). 1
 - c. After what time period will the number of bacteria have doubled? 1
 - d. At what rate is the number of bacteria increasing when $t = 30$ seconds? 1

Question 16 Start this question in a new booklet.

α)



- | | | |
|------|--|---|
| i. | Find the gradient of AB. | 1 |
| ii. | Find the midpoint of AB. | 1 |
| iii. | Show that the equation of the perpendicular bisector of AB is $14x - 4y + 15 = 0$. | 1 |
| iv. | What is the perpendicular distance of the line found in part iii to the origin (0, 0)? | 1 |

β)

- | | | |
|------|---|---|
| i. | An arithmetic series has first term 3 and common difference 4. | 3 |
| | a. Find the 17 th term. | |
| | b. Find the sum of the tenth to the twentieth terms (inclusive). | |
| ii. | The sum of the first 8 terms of a geometric series is 17 times the sum of its first 4 terms. Find the common ratio. | 2 |
| iii. | Colin borrows \$30 000 to buy a new car. He is charged an interest rate of 15% pa, calculated and compounded monthly, and an amount M is repaid every month. If A_n is the amount owing after n months, show that | |
| a. | $A_2 = 30\,000 \times 1.0125^2 - M(1 + 1.0125)$ | 1 |
| b. | Hence, show that $A_n = \$30\,000 \times 1.0125^n - M\left(\frac{1.0125^n - 1}{0.0125}\right)$ | 1 |
| c. | Find the value of M, to the nearest cent, if the loan is repaid at the end of 7 years. | 2 |
| d. | How much extra, in total, will be repaid if the loan is taken over 10 years? (Answer to the nearest dollar) | 2 |

2013 TRIAL MATHEMATICS

SOLUTIONS

QUESTION 11

a) i) $5.9 \checkmark$

(1)

ii) $5\sqrt{3} + \sqrt{20} - 2\sqrt{12}$

$5\sqrt{3} + \sqrt{4 \times 5} - 2\sqrt{4 \times 3}$

$5\sqrt{3} + 2\sqrt{5} - 4\sqrt{3} \checkmark$

$= \sqrt{3} + 2\sqrt{5} \checkmark$ (2)

iii) $3x^2 + 8x - 16$

$\begin{array}{r} 3x \quad -4 \\ \times \quad \quad 4 \\ \hline \end{array}$

$(3x-4)(x+4)$ (2)

iv) $|3n-5| = 4n-7$

$3n-5 = 4n-7 \quad 5-3n = 4n-7$

$2 = n$

$-7n = -12$

$(1=1) \checkmark$

$n = 12/7$

$n=2$ is a solution

$(-31 \neq 4) \checkmark$

$n=12$ not sol'n.

(2)

v) $\frac{3x-4}{2} = \frac{x+1}{5}$

$\frac{15x-20}{10} = \frac{(2x+2)}{10}$

$15x-20 = 2x+2$

$13x-22 = 2$ $\checkmark$

10

$13x-22 = 2$ $\checkmark$

10

(2)

β) i) a) $\alpha + \beta = -\frac{b}{a} = \frac{6}{3} = 2 \checkmark$ (1)

b) $x^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$\alpha\beta = \frac{12}{3} = 4 \checkmark$

$x^2 + \beta^2 = 2^2 - 2(4)$ (2)

$= 4 - 8$

$= -4 \checkmark$

(1)

Q1-10 at end

ii) $x^2 + (m-3)x + m = 0$

a) $\Delta = b^2 - 4ac$

$= (m-3)^2 - 4(1)(m)$

$= m^2 - 6m + 9 - 4m$

$= m^2 - 10m + 9$

$\begin{array}{r} m \quad -9 \\ \times \quad \quad -1 \\ \hline \end{array}$

$m \quad -1$

$= (m-9)(m-1) \checkmark$ (1)

b) $\Delta > 0$ has two unequal roots

$\Delta > 0$

$(m-9)(m-1) > 0$

$m = 9, m = 1$

$m > 9 \checkmark$

$m < 1 \checkmark$

(2)

to have two real distinct roots.

12. i) $\cos \frac{\pi}{6} = \left[\frac{\sqrt{3}}{2} \right]$

(ii) $5\cos^2 \theta + 2\sin \theta - 2 = 0, \quad 0 \leq \theta \leq 2\pi$

$$5(1 - \sin^2 \theta) + 2\sin \theta - 2 = 0$$

$$-5\sin^2 \theta + 2\sin \theta + 3 = 0$$

$$5\sin^2 \theta - 2\sin \theta - 3 = 0$$

Let $x = \sin \theta$ $(5x - 5)(5x + 3) = 0$ $\begin{matrix} \times -15 \\ + -2 \\ \hline -5.3 \end{matrix}$

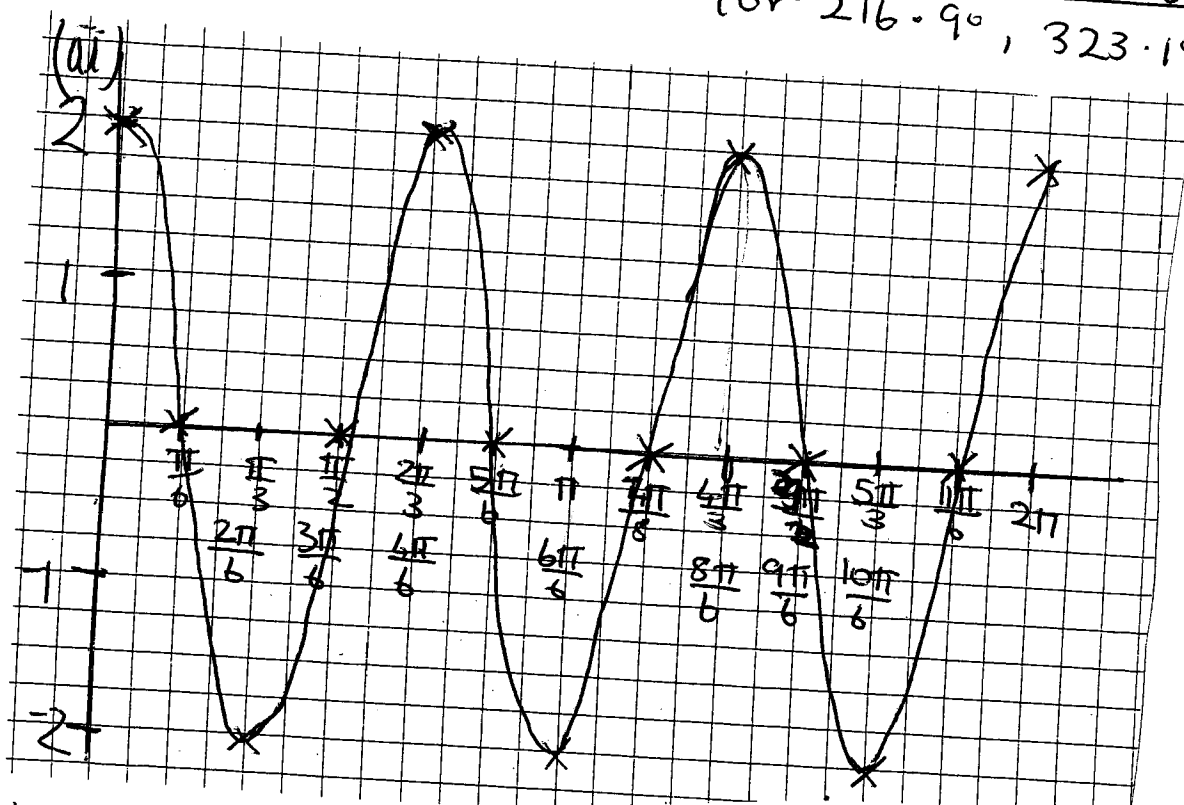
$x = 1$ or $-3/5$

$\sin \theta = 1$

$\therefore \theta = \frac{\pi}{2}$ (or 1.570)

$\sin \theta = -3/5$

$\therefore \theta = 3.785$ or 5.640 (to 4 sf)
(or $216.9^\circ, 323.1^\circ$)



Period = $\frac{2\pi}{3}$
Amplitude = 2

(IV) $\frac{d}{dx}(\sin 2x) = \underline{2\cos 2x}$

(V)
$$\begin{aligned} \text{LHS} &= \frac{\sin \theta \cos^2 \theta - \sin^3 \theta}{2\cos^3 \theta - \cos \theta} = \frac{\sin \theta (\cos^2 \theta - \sin^2 \theta)}{\cos \theta (2\cos^2 \theta - 1)} \\ &= \frac{\sin \theta (\cos^2 \theta - (1 - \cos^2 \theta))}{\cos \theta (2\cos^2 \theta - 1)} = \frac{\sin \theta (2\cos^2 \theta - 1)}{\cos \theta (2\cos^2 \theta - 1)} \\ &= \frac{\sin \theta}{\cos \theta} = \tan \theta = \text{RHS} \end{aligned}$$

(2)

$$12 \text{ vi) a) } A = \frac{1}{2} r^2 \theta$$

$$90\pi = \frac{1}{2} (15)^2 \theta$$

$$\theta = \frac{180\pi}{225} = \boxed{\frac{4\pi}{5}}$$

$$\text{b) } L = r\theta$$

$$= 15 \times \frac{4\pi}{5} = \underline{12\pi}$$

$$AC^2 = 15^2 + 15^2 - 2 \times 15 \times 15 \cos \frac{4\pi}{5}$$

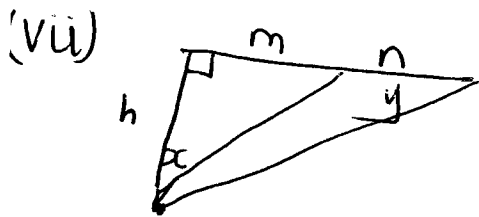
$$= 450 - 450 \cos \frac{4\pi}{5}$$

$$= 814.05 \dots$$

$$\underline{AC = 28.53 \dots}$$

$$P = AC + 12\pi$$

$$\underline{P = 66 \text{ cm (to nearest cm)}}$$



$$\frac{\tan x}{\text{(SOHCAHTOA)}} = \frac{m}{h} \left(\frac{\text{opp}}{\text{adj}} \right) \quad \tan y = \frac{h}{m+n}$$

$$\tan x \cdot \tan y = \frac{m}{h} \times \frac{h}{m+n}$$

$$= \boxed{\frac{m}{m+n}}$$

Q13

i) $\frac{dy}{dx} = 6x^2 - 6$

$$y = 2x^3 - 6x + C$$

$(2, -1)$

$$-1 = 2(2)^3 - 6(2) + C$$

$$-1 = 16 - 12 + C$$

$$-5 = C$$

$$\therefore y = 2x^3 - 6x - 5$$

ii)

x	0	1	2	3	4
y	0.2	0.25	0.3	0.5	1

$$A = \frac{1}{3} [0.2 + 4(0.25 + 0.5) + 2(0.3) + 1]$$

$$= \frac{1}{3} [1.2 + 3 + 0.6]$$

$$= \frac{1}{3} [4.8]$$

$$= 1.6$$

iii) a) $A = \frac{1}{4}\pi + \frac{1}{2} + 3$

$$= \frac{\pi + 2 + 12}{4}$$

$$= \frac{\pi + 14}{4}$$

b) $\int_0^5 f(x) dx = -\frac{1}{4}\pi + \frac{1}{2} - 3$

$$= \frac{-\pi + 2 - 12}{4}$$

$$= \frac{-10 - \pi}{4}$$

iv) a) $y = -x^2 + 1$ --- ①

$$y = \frac{3x}{2}$$
 --- ②

$$\therefore \frac{3x}{2} = -x^2 + 1$$

$$x^2 + \frac{3x}{2} - 1 = 0$$

$$2x^2 + 3x - 2 = 0$$

$$\frac{(2x+4)(2x-1)}{2}$$

$$\therefore (x+2)(2x-1) = 0$$

$$\therefore x = -2 \text{ or } x = \frac{1}{2}$$

B is in 1st Quad.

$$\therefore B\left(\frac{1}{2}, \frac{3}{4}\right)$$

a) $y = -x^2 + 1$

$$\therefore A(0, 1)$$

c) $V = \pi \int_{\frac{3}{4}}^1 x_1^2 dy + \pi \int_0^{\frac{3}{4}} x_2^2 dy$

$$y = -x^2 + 1, \quad y = \frac{3x}{2}$$

$$x^2 = 1 - y, \quad x = \frac{2y}{3}$$

$$x^2 = \frac{4y^2}{9}$$

$$= \pi \int_{\frac{3}{4}}^1 (1-y) dy + \pi \int_0^{\frac{3}{4}} \frac{4y^2}{9} dy$$

$$= \pi \left[y - \frac{y^2}{2} \right]_{\frac{3}{4}}^1 + \frac{4\pi}{9} \left[\frac{y^3}{3} \right]_0^{\frac{3}{4}}$$

$$= \pi \left[\left(1 - \frac{1}{2}\right) - \left(\frac{3}{4} - \frac{9}{32}\right) \right] + \frac{4\pi}{27} \left[\frac{27}{64} \right]$$

$$= \pi \left[\left(\frac{1}{2}\right) - \left(\frac{3}{4} - \frac{9}{32}\right) \right] + \frac{\pi}{16}$$

$$= \frac{\pi}{32} + \frac{\pi}{16}$$

(4)

2 U - Trial 2013
Qn(14) - Solution

$$\boxed{i} \quad 3 \log_7 2 = \log_7 x - \log_7 4$$

$$\log_7 8 = \log_7 \frac{x}{4}$$

$$8 = \frac{x}{4} \Rightarrow x = 32$$

$$\boxed{ii} \quad a) \quad y = x^2 \ln x$$

$$y' = x^2 \cdot \frac{1}{x} + 2x \ln x$$

$$= x + 2x \ln x$$

$$= x(2 \ln x + 1)$$

$$b) \quad y = \frac{e^x}{3x-2}$$

$$y' = \frac{(3x-2)e^x - 3e^x}{(3x-2)^2}$$

$$\boxed{iii} \quad \int 5e^{2x} dx = \frac{5}{2}e^{2x} + C$$

$$\boxed{iv} \quad \int \frac{4}{2x-7} dx = 2 \ln(2x-7) + C$$

$$\boxed{v} \quad \int_0^1 x e^{x^2} dx = \frac{1}{2} \int_0^1 2x e^{x^2} dx$$

$$= \frac{1}{2} [e^{x^2}]_0^1$$

$$= \frac{1}{2}(e-1)$$

$$\boxed{vi}$$

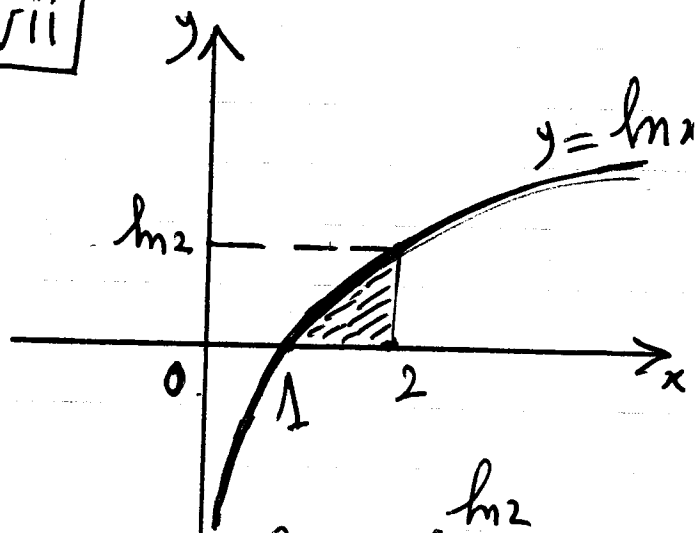
$$y = 2xe^x$$

$$y' = 2xe^x + 2e^x, x=1$$

$$m = 4e$$

$$y-e = 4e(x-1)$$

$$\boxed{vii}$$



$$\text{Area} = 2 \ln 2 - \int_0^{\ln 2} x dy$$

$$= 2 \ln 2 - \int_0^{\ln 2} e^y dy$$

$$= 2 \ln 2 - [e^y]_0^{\ln 2} = 2 \ln 2 - 1$$

$$15. x \quad y = x^3 - 3x^2 - 9x + 1$$

$$i. \frac{dy}{dx} = 3x^2 - 6x - 9$$

$$ii. \frac{dy}{dx} = 0$$

$$3(x^2 - 2x - 3) = 0$$

$$(x-3)(x+1) = 0$$

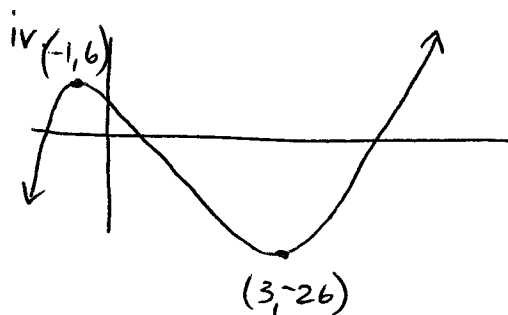
$$x = 3, -1$$

$$(3, -26) \quad (-1, 6)$$

$$iii. \frac{d^2y}{dx^2} = 6x - 6$$

$$(3, -26) \quad \frac{d^2y}{dx^2} = 12 > 0 \quad \uparrow \uparrow \quad \text{min}$$

$$(-1, 6) \quad \frac{d^2y}{dx^2} = -12 < 0 \quad \downarrow \downarrow \quad \text{max}$$



$$B. x = t + \frac{25}{t+2}$$

$$a. t=0 \quad x = 12\frac{1}{2} \text{ metres}$$

$$b. \dot{x} = 1 - 25(t+2)^{-2}$$

$$= 1 - \frac{25}{(t+2)^2}$$

$$\ddot{x} = 50(t+2)^{-3}$$

$$= \frac{50}{(t+2)^3}$$

$$c. \text{rest } \dot{x} = 0$$

$$0 = 1 - \frac{25}{(t+2)^2}$$

$$1 = \frac{25}{(t+2)^2}$$

$$t^2 + 4t + 4 = 25$$

$$t^2 + 4t - 21 = 0$$

$$(t+7)(t-3) = 0$$

$$t = 3, -7 \quad \therefore t = 3.$$

$$x = 3 + \frac{25}{5} = 8 \text{ metres.}$$

$$d. \dot{x} = 1$$

$$ii. N = 25000e^{0.007t}$$

$$a. t=0 \quad N = 25000$$

$$b. N = 25000e^{0.007 \times 30}$$

$$= 30841.95$$

$$= 30842$$

$$c. 50000 = 25000e^{0.007t}$$

$$2 = e^{0.007t}$$

$$t = \frac{\ln 2}{0.007} = 99.02 \text{ sec}$$

$$d. \frac{dN}{dt} = 0.007 \times 25000e^{0.007 \times 30}$$

$$= 0.007 \times 30841.95$$

$$= 215.89 \text{ bacteria / second}$$

$$16) \alpha) i) m_{AB} = \frac{1-3}{3+4} = -\frac{2}{7} \quad (1)$$

$$ii) M_{AB} = \left(-\frac{4+3}{2}, \frac{3+1}{2} \right) \\ = \left(-\frac{7}{2}, 2 \right) \quad (1)$$

$$iii) y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{7}{2} \left(x + \frac{1}{2} \right)$$

$$2y - 4 = 7x + \frac{7}{2}$$

$$4y - 8 = 14x + 7 \\ 14x - 4y + 15 = 0 \quad (1)$$

$$iv) d_1 = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \quad (0,0)$$

$$= \frac{|14(0) - 4(0) + 15|}{\sqrt{14^2 + (-4)^2}}$$

$$\therefore d_1 = \frac{15}{\sqrt{212}} \quad \text{or} \quad \frac{15}{2\sqrt{53}} \\ \text{or} \quad \frac{15\sqrt{53}}{106} \\ \text{or} \approx 1.03 \text{ units} \quad (1)$$

$$\beta) i) a) a=3, d=4$$

$$T_n = a + (n-1)d$$

$$T_{17} = 3 + (16)4 \\ = 67 \quad (1)$$

$$b) S_{20} - S_{10} + T_{10}$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{10} = 5(6 + (9)4) \\ = 210$$

$$S_{20} = 10(6 + (19)4) \\ = 820$$

$$T_{10} = 3 + (9)4 \\ = 39$$

$$\therefore \text{sum} = 820 - 210 + 39 \quad \checkmark \quad (2) \\ = 649$$

$$16 \beta ii) S_8 = 17 S_4$$

$$\frac{r^8 - 1}{r - 1} = \frac{17(r^4 - 1)}{r - 1}$$

$$r^8 - 1 = 17(r^4 - 1) \\ (\cancel{r^4 - 1})(r^4 + 1) = 17(\cancel{r^4 - 1})$$

$$r^4 + 1 = 17$$

$$r^4 = 16$$

$$r = \pm 2$$

$$\therefore r = 2$$

1b) iii) a) $A_0 = \$30,000$; $r = 1.0125$
 $A_1 = 30000 \times 1.0125 - m$
 $A_2 = A_1 \times 1.0125 - m$
 $= (30000 \times 1.0125 - m) \times 1.0125 - m$
 $= 30000 \times 1.0125^2 - 1.0125m - m$
 $\therefore A_2 = 30000 \times 1.0125^2 - m(1 + 1.0125)$ (1)

b) $A_3 = A_2 \times 1.0125 - m$
 $= [30000 \times 1.0125^2 - m(1 + 1.0125)] \times 1.0125 - m$
 $= 30000 \times 1.0125^3 - m(1 + 1.0125 + 1.0125^2)$

$\therefore A_n = 30000 \times 1.0125^n - m(1 + 1.0125 + 1.0125^2 + \dots + 1.0125^{n-1})$

with common ratio 1.0125
 and $Sum = \frac{1.0125^n - 1}{0.0125}$

Hence $A_n = 30000 \times 1.0125^n - m \left(\frac{1.0125^n - 1}{0.0125} \right)$ (1)

c) $30000 \times 1.0125^{84} - m \left(\frac{1.0125^{84} - 1}{0.0125} \right)$ ✓
 $m = \frac{0.0125 \times 30000 \times 1.0125^{84}}{1.0125^{84} - 1}$
 $A_{84} = \$578.90$ ✓ (2)

d) $84 \times \$578.90 = \48627.82 ✓

$A_{120} = 30000 \times 1.0125^{120} - m \left(\frac{1.0125^{120} - 1}{0.0125} \right)$
 $= \$58080.58$ ✓

$\therefore \text{extra paid} = \$58080.58 - \$48627.82$

$= \$9452.76$ (2)

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct
 ↗

ANSWERS

Start
Here →

1. A ☒ B ☐ C ☐ D ☐
2. A ☐ B ☒ C ☐ D ☐
3. A ☒ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☒
5. A ☐ B ☒ C ☐ D ☐
6. A ☐ B ☐ C ☒ D ☐
7. A ☐ B ☐ C ☐ D ☒
8. A ☐ B ☐ C ☐ D ☒
9. A ☐ B ☐ C ☐ D ☒
10. A ☐ B ☐ C ☒ D ☐

Section 1